Group 3 - Space Complexity

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Implementation

Libraries and assets used:

- LibGDX (Library) - Uses Apache License 2.0, a permissive open-source licence. This was useful to us as it meant we could use it freely and treat it as our own game.
- GSON (Library) - Uses an Apache 2.0 licence, allowing us to use it freely.
- Glassy UI (Asset) (https://github.com/czyzby/gdx-skins/tree/master/glassy) - Uses CC BY 4.0 Licence, this licence allows usage, modification and sharing of the model for both commercial and non-commercial purposes, as long as users follow specific guidelines, like providing credit, this is useful as we can use it freely within our game.
- DALL-E - Uses a Custom Policy by OpenAI, this licence allows us to use these images for commercial, personal, or non-commercial purposes meaning we can use the generated images freely within our game.

All other assets were created by our team or the previous team, hence we do not need to worry about copyright issues associated with online assets. This also gave us more creative freedom as we could design assets that would better suit our game, as opposed to having to tailor the game to found assets.

We also used tools as-is to test our game in order to efficiently find issues and resolve them:

- JUnit (Library) - Used in our automated testing to create unit tests. Uses Apache License 2.0 allowing us to freely use it as-is to help develop our game.
- Mockito (Library) - Used in our tests to mock certain classes. Uses MIT License allowing us to freely use it as-is to help develop our game.
- Jacoco (Plugin) - Used in our testing to generate coverage reports. Uses Eclipse Public License Version 2.0 allowing us to freely use it as-is to help develop our game.
- Checkstyle (Plugin) - Used in our automated testing to check code style. Uses GNU LGPL licence v2.1 allowing us to freely use it as-is to help develop our game.

Improvements have been made to previously implemented features, as well as implementing the new requirements given to us after taking over the project.

- The leaderboard that was previously planned and unimplemented due to time restraints, has now been completed and added to the game with full functionality.
- The game config data that was previously saved to a binary file has now been improved. Being saved as JSON in the game prefs file.

All of the requirements have been met include:

- FR_MAP - The game provides a map large enough for the player to fulfil objectives.
- FR_ACCOMODATION_BUILDING - At least 2 placeable accommodation buildings in the game.
- FR_LEARNING_BUILDING - At least 2 placeable learning buildings in the game.
- FR_EATING_BUILDING - At least 2 placeable buildings for students to eat in the game.
- FR_RECREATIONAL_BUILDING - At least 2 placeable recreational buildings in the game.
- FR_PLANNED_EVENTS - Include predictable planned events such as graduation.
- FR_UNPLANNED_EVENTS - Include unpredictable random events that the player has to react to and handle appropriately.
- FR_LEADERBOARD - Include a leaderboard with names and scores of the top 5 users.
- FR_DISPLAY_ACHIEVEMENTS - Show player achievements they have received.
- FR_MEASURE_ACHIEVEMENTS - Measure player progress in achievements.